Innovation Centre for Education

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Yenepoya Institute Of Arts Science Commerce and MAnagenent

Project – High Level Design (HLD)

PG Recommendation System for Smart Cities

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1. Introduction

1.1 Scope of the document

This document provides a high-level overview of the PG Recommendation System. It describes system architecture, components, and overall workflow without going into implementation details.

1.2 Intended Audience

- Project evaluators
- Faculty members
- Developers

1.3 System Overview

The PG Recommendation System is designed to help users find suitable accommodation in urban areas based on:

- City
- Budget
- Tenant type

The system processes a dataset and provides filtered recommendations to improve decision-making in smart cities.

2. System Design

2.1 Application Design

Architecture Type:

- Modular Python-based application
- Rule-based recommendation system (Phase 1)

Development Environment:

- Visual Studio Code (VS Code)

Technology Stack:

- Python
- Pandas

Future Enhancements:

- Machine Learning integration
- Web-based interface (Flask/Streamlit)

2.2 Process Flow

1. User provides input (city, budget, tenant type)
2. Input module captures data
3. Data module loads dataset
4. Recommendation engine processes data
5. Filtered results are generated
6. Output module displays recommendations

2.3 Information Flow

User → Input Interface → Recommendation Engine → Filtered Data → Output Display

2.4 Component Design

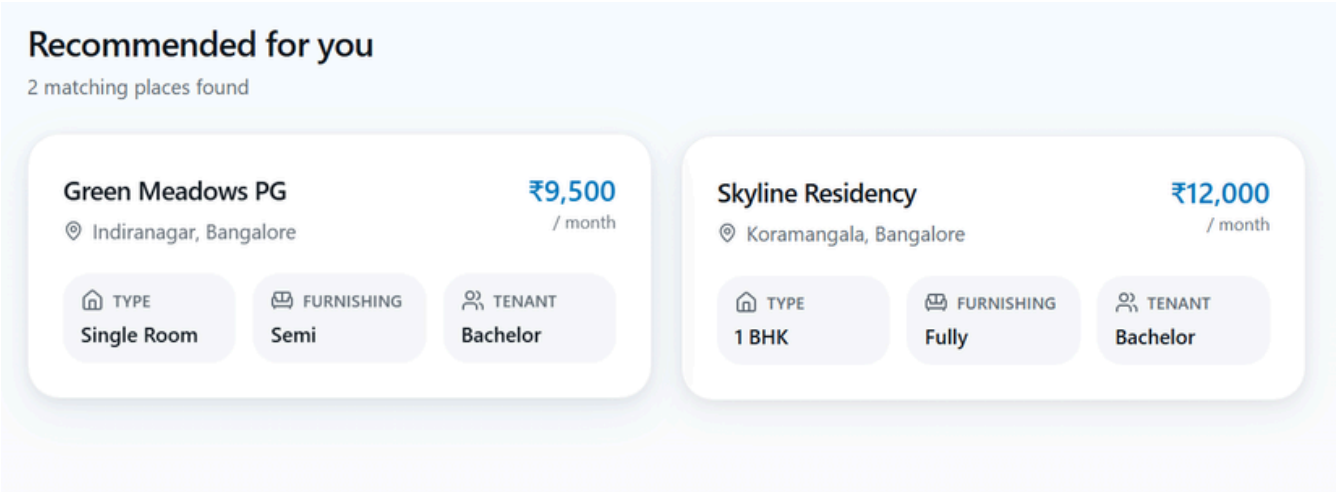
1. Input Module
   - Collects user preferences via CLI
   - Validates input

2. **Data Management Module**

- Loads dataset from CSV
- Handles data preprocessing

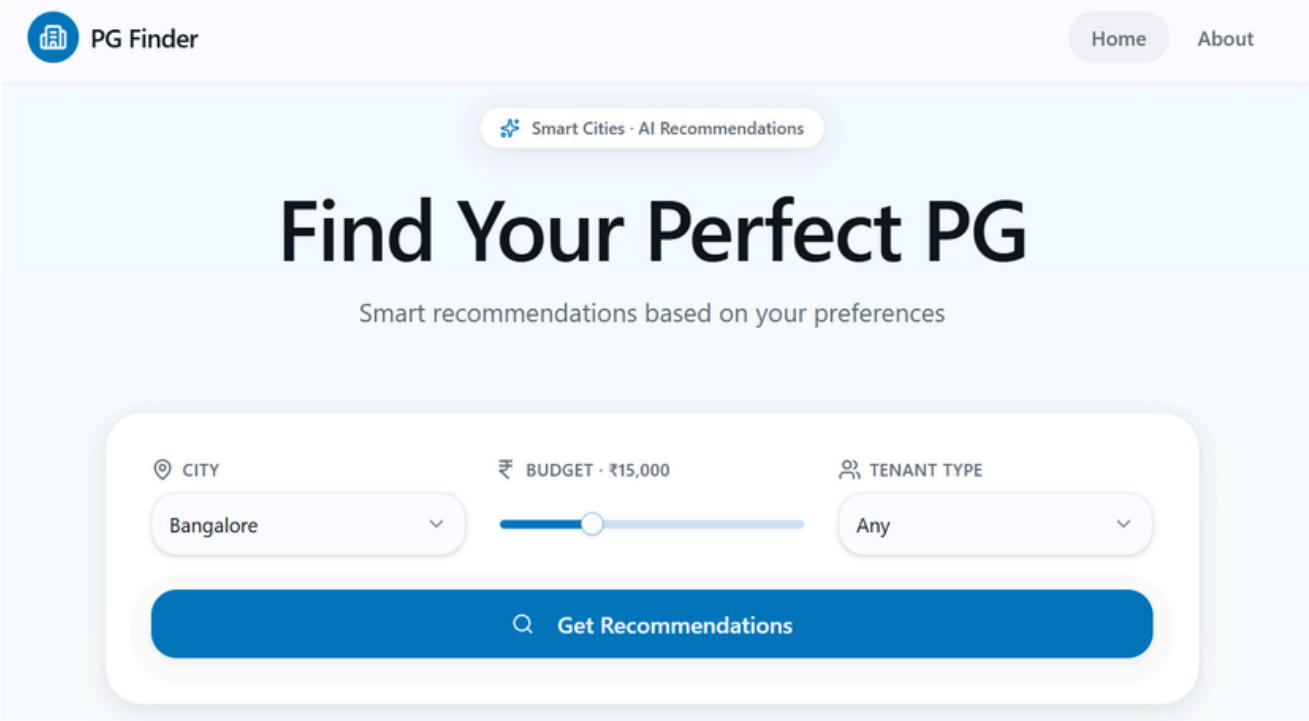
3. **Recommendation Engine**

- Core logic layer
- Applies filtering conditions
- Matches user preferences



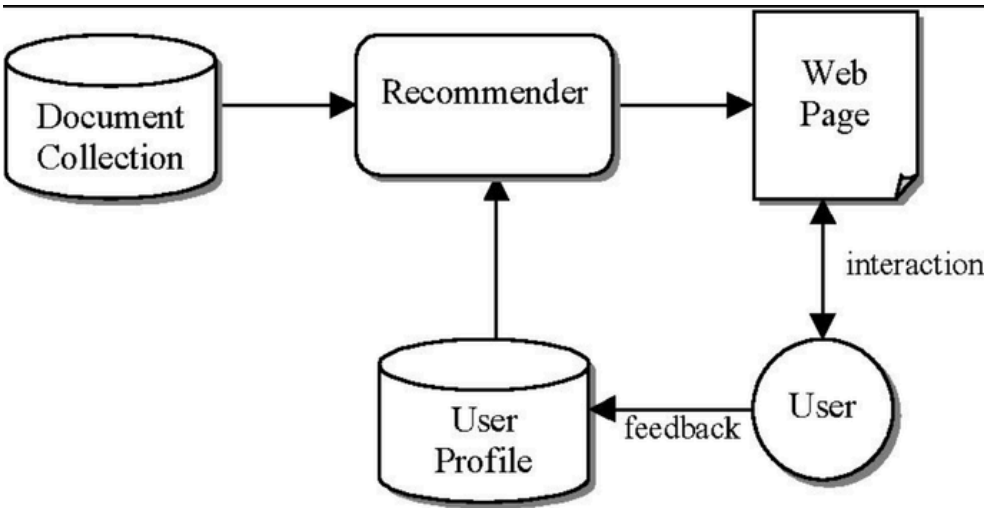
4. **Output Module**

- Displays formatted results
- Ensures readability



This is a minimal UI designed to collect user preferences and display PG recommendations in a clean, card-based format.

- Minimal and professional interface
- Clean layout with soft colors and modern typography
- Focus on usability and clarity



2.5 Key Design Considerations

- Modular and scalable design
- Easy integration with ML models
- Efficient data filtering
- Maintainability

2.6 API Catalogue

- Not implemented (current version)
- Future: REST APIs using Flask

3. Data Design

3.1 Data Model

Fields:

- City
- Rent
- BHK
- Furnishing Status
- Tenant Preferred
- Location

3.2 Data Access Mechanism

- CSV-based storage
- Access via pandas

3.3 Data Retention Policies

- Static dataset
- No user data stored

3.4 Data Migration

Not required (initial version)

4. Interfaces

- Command-line interface (CLI)
- User interacts via terminal

5. State and Session Management

- No session handling
- Stateless system

6. Caching

Not implemented

7. Non-Functional Requirements

7.1 Security Aspects

No sensitive data used

7.2 Performance Aspects

- Fast filtering using pandas
- Suitable for small to medium datasets

8. References

- Python Documentation
- Pandas Documentation